Psychological statistics

Homework Assignment ③

B812109041 Isaac, Lee

1. ESSAY GRADE

```
Residuals:
     Min       1Q   Median       3Q      Max 
-13.0407  -4.1877  -0.7728   4.7592  16.7620 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  57.9296     3.1970  18.120   <2e-16 ***
hours         0.6612     0.3644   1.814   0.0766 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.588 on 43 degrees of freedom
Multiple R-squared:  0.07112,	Adjusted R-squared:  0.04952 
F-statistic: 3.292 on 1 and 43 DF,  p-value: 0.07659
```

Adjusted R – squared: 0.04952.

Conclusion:

The fitness of the linear model is relatively low, and the positive correlation is less evident.

2. CHECK THE PAPER

I. Calculate the correlation coefficient between CAQ scores and AUT fluency.

```
Call:
lm(formula = CAQ.scores ~ Fluency)

Residuals:
    Min      1Q  Median      3Q     Max 
-12.858  -5.375  -3.094   1.845  48.675 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.775      5.347   0.519    0.607
Fluency        2.573      1.644   1.565    0.126

Residual standard error: 10.25 on 38 degrees of freedom
Multiple R-squared:  0.06055,	Adjusted R-squared:  0.03583 
F-statistic: 2.449 on 1 and 38 DF,  p-value: 0.1259
```

Adjusted R – squared: 0.03583 / p-value: 0.1259.

II. Calculate the correlation coefficient between CAQ scores and AUT originality.

```
Call:
lm(formula = CAQ.scores ~ Origenality)

Residuals:
    Min      1Q  Median      3Q     Max 
-10.658  -6.717  -2.331   2.353  44.438 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)    1.593      4.897   0.325   0.7467  
Origenality   27.917     14.120   1.977   0.0553 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.07 on 38 degrees of freedom
Multiple R-squared:  0.09328,	Adjusted R-squared:  0.06942 
F-statistic: 3.909 on 1 and 38 DF,  p-value: 0.05531
```

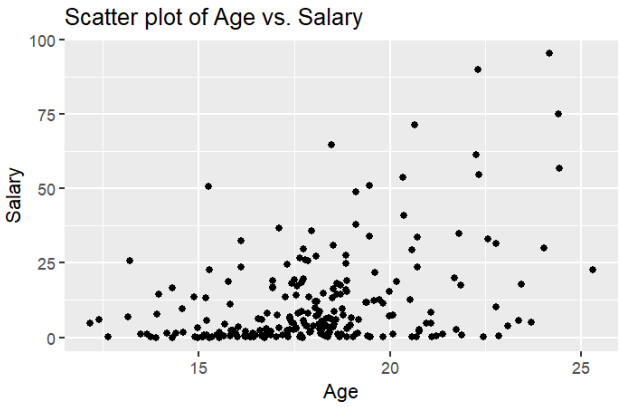
Adjusted R – squared: 0.06942 / p-value: 0.055331.

III. Compare the CAQ scores between the high creativity group and the low creativity group.

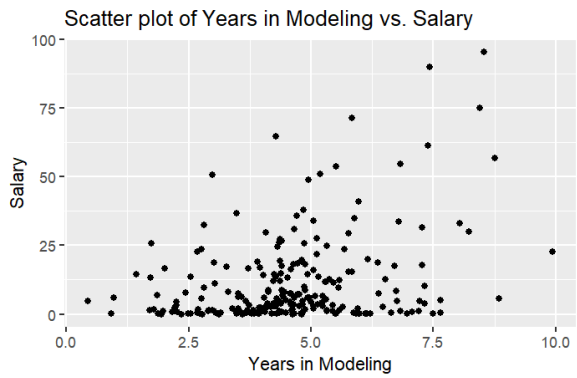
I used t.test to compare the two groups, but I couldn't successfully divide the data frame into high-score group and low-score group.

3. SUPERMODEL

```
Call:
lm(formula = SALARY ~ AGE + YEARS + BEAUTY, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-24.853  -7.950  -4.197   4.605  68.085 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -60.8897    16.4966  -3.691  0.00028 ***
AGE           6.2344     1.4112   4.418 1.54e-05 ***
YEARS        -5.5612     2.1222  -2.621  0.00937 ** 
BEAUTY       -0.1964     0.1524  -1.289  0.19871    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 14.57 on 227 degrees of freedom
Multiple R-squared:  0.184,	Adjusted R-squared:  0.1733 
F-statistic: 17.07 on 3 and 227 DF,  p-value: 4.973e-10
```



Scatter plot of Age vs. Salary



Scatter plot of Years in Modeling vs. Salary

Conclusion:

I. Age has a significant positive (coefficient = 6.2344, p = 0.00028) effect on salary, meaning that as age increases, salary also increases accordingly. On the other hand, the number of years spent in the modeling profession has a significant negative effect (coefficient = -5.5612, p = 1.54e-05) on salary.

II. The attractiveness rating, however, does not have a statistically significant effect on salary prediction. (R-square = 0.1733)

4. CHILD AGGRESSION

```
Call:
lm(formula = Aggression ~ Parenting_Style + Television + Diet +
    Computer_Games + Sibling_Aggression, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.12629 -0.15253 -0.00421  0.15222  1.17669

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        -0.004988   0.011983  -0.416 0.677350
Parenting_Style     0.056648   0.014557   3.891 0.000110 ***
Television          0.032916   0.046057   0.715 0.475059
Diet               -0.109054   0.038076  -2.864 0.004315 **
Computer_Games      0.142161   0.036920   3.851 0.000129 ***
Sibling_Aggression  0.081684   0.038780   2.106 0.035550 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3071 on 660 degrees of freedom
Multiple R-squared:  0.08258,	Adjusted R-squared:  0.07563
F-statistic: 11.88 on 5 and 660 DF,  p-value: 5.025e-11
```

According to the regression analysis results:

1. **Interpretation of Coefficients**

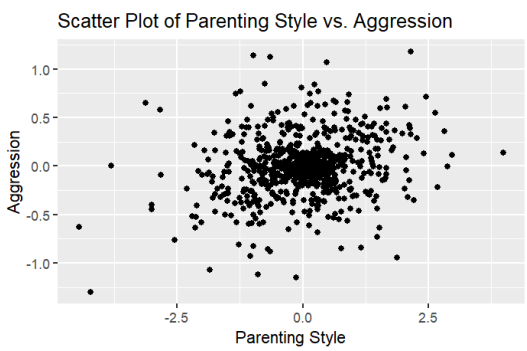
   I. The coefficient for Parenting Style is 0.0566, with a standard error of 0.0146 and a significant p-value (< 0.001), indicating a significant positive effect on aggression.

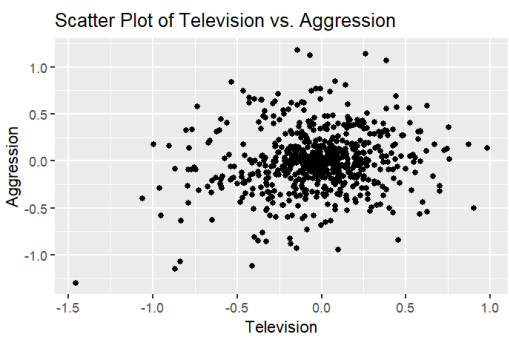
   II. Television and Sibling Aggression show some influence but are not statistically significant.

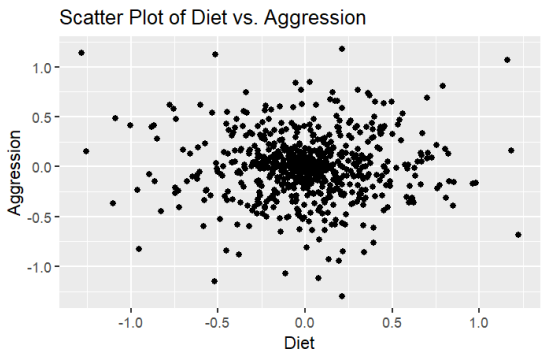
   III. Diet and Computer Games have a significant negative impact on aggression, meaning that more time spent on diet and computer games is associated with less aggression.

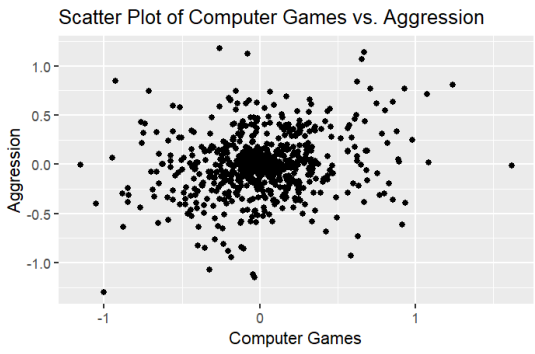
2. Model Fit

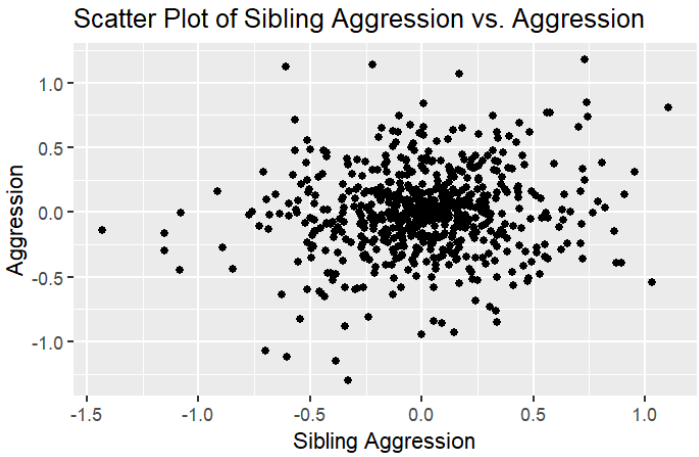
   I. The model's R-squared is 0.08258, indicating that it can explain about 8.26% of the variance in aggression, suggesting a relatively low overall explanatory power.











Conclusion

Overall, these results suggest that Parenting Style, Diet, and Computer Games may significantly influence children's aggression, while Television viewing and Sibling Aggression have a lesser, or even insignificant, impact. This supports previous research indicating that Parenting Style and Sibling Aggression are important predictors of children's aggression.